THE NATIONAL UNIVERSITY OF LESOTHO
Department of Mathematics and Computer Science — Faculty of Science and Technology

M3503 — Ordinary Differential Equations

Unit 1: First-Order Ordinary Differential Equations

A revision of solution methods, with areas of application

1. Introduction: first-order equations and the art of recognition

Almost every quantitative law in science begins life as a statement about a *rate* — how fast a quantity grows, cools, decays, spreads or flows — long before anyone can write down the quantity itself. A first-order differential equation is precisely the sentence that converts such a rate law back into behaviour over time. This is why the course begins here: the first-order equation is the smallest complete object in the theory, and the great majority of applied models are either first-order or assembled from first-order parts.

You met these methods in M2502, and this unit is a deliberate revision — but not a hurried parade of tricks. The single most valuable habit in the whole course is **recognition**: looking at an unfamiliar equation and knowing, within a few seconds, which of a small number of standard forms it belongs to, and therefore which tool will open it. Each section below is built as the answer to one recognition question, and the unit closes with a guide (§7) that drills the instinct directly. Master it, and the later units — higher-order equations, systems, series solutions — become variations on a theme you already own.

Keep **three lenses** in view throughout. Any equation can be approached *analytically* (an exact formula — the business of this unit), *qualitatively* (its equilibria, stability and long-run behaviour, read straight from the equation without solving), or *numerically* (approximated step-by-step on a computer). Unit 1 lives mostly in the first lens; but the same equations return under the other two throughout the course, and it is worth knowing from the outset that "no elementary formula exists" is a routine outcome, not a failure.

How to read this unit. Each method is presented the same way: a definition that tells you how to *recognise* the form; a short account of *why* its technique works, not merely the steps; a boxed recipe; a fully worked example; and a note on where the form arises in science and engineering. A recognition guide (§7) and a self-check problem set (§8) close the unit.

Aligns with course outcomes H1 (analyse and solve IVPs), H2 (compare analytical methods) and H5 (apply to real-life problems). Reference texts: Nagle, Saff & Snider, Ch. 2; Zill, Ch. 2.

What it means to solve a first-order equation

A **first-order ODE** is any equation relating x, an unknown function $y(x)$, and its first derivative:

$$F\left(x,\, y,\, \frac{dy}{dx}\right) = 0, \qquad \text{most often written} \qquad \frac{dy}{dx} = f(x, y).$$

A **solution** on an interval I is a differentiable function $y(x)$ that makes the equation an identity on I. Because integration introduces an arbitrary constant, a first-order equation carries a one-parameter family of solutions — its **general solution**. Prescribing one value $y(x_0) = y_0$ (an

initial condition) singles out a **particular solution**; the pair

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0$$

is an **initial value problem** (IVP).

Theorem 1.1 (Existence & Uniqueness — Picard–Lindelöf)**.** *If $f(x, y)$ and $\partial f/\partial y$ are continuous on a rectangle containing (x_0, y_0), then the IVP has a unique solution on some interval about x_0.*

This theorem is our licence: when its hypotheses hold, the trajectory through a point exists and does not branch, so the answer we compute is *the* answer. We now build the toolkit.

2. Separable Equations

Definition 2.1. An equation is **separable** if the right-hand side factors into a part depending only on x times a part depending only on y:

$$\frac{dy}{dx} = g(x)\, h(y).$$

Why the method works. Treat $\dfrac{dy}{dx}$ as a ratio of differentials (justified by the chain rule). Collect all y's on one side and all x's on the other, then integrate both sides against their own variable:

$$\frac{dy}{h(y)} = g(x)\, dx \implies \int \frac{dy}{h(y)} = \int g(x)\, dx + C.$$

Each integral is now an ordinary single-variable integral. The constant C carries the one-parameter family.

Recipe

(1) Write $\dfrac{dy}{dx} = g(x)h(y)$. (2) Separate: $\dfrac{dy}{h(y)} = g(x)\, dx$. (3) Integrate both sides. (4) Apply any initial condition to fix C. (5) **Check for lost solutions**: any constant $y = c$ with $h(c) = 0$ is an *equilibrium solution* that division by $h(y)$ may have discarded.

Example 2.1 (Growth and decay)**.** Solve $\dfrac{dP}{dt} = kP$. Separate and integrate: $\displaystyle\int \frac{dP}{P} = \int k\, dt \Rightarrow$ $\ln|P| = kt + C \Rightarrow P(t) = P_0 e^{kt}$, where $P_0 = P(0)$. Positive k gives growth, negative k gives decay. The constant solution $P \equiv 0$ (where $h(P) = P$ vanishes) is the equilibrium.

Example 2.2 (A geometric family)**.** Solve $\dfrac{dy}{dx} = -\dfrac{x}{y}$. Then $y\, dy = -x\, dx$, so $\frac{1}{2}y^2 = -\frac{1}{2}x^2 + C_1$, giving the concentric circles $x^2 + y^2 = C$. The equation says the solution curve is everywhere perpendicular to the radius — which is exactly the geometry of a circle.

Where this shows up

- **Radioactive decay & carbon dating:** $\dot{N} = -\lambda N$ gives the half-life law used to date artefacts and rock.
- **Newton's law of cooling:** $\dot{T} = -k(T - T_{\text{room}})$ predicts how fast a body reaches ambient temperature.
- **Torricelli's law:** a draining tank obeys $\dot{h} = -c\sqrt{h}$ — separable, and the basis of water-

clock and outflow design.

- **First-order reaction kinetics:** concentration in a chemical reaction, $\dot{c} = -kc$.
- **Unconstrained population / compound interest:** the $\dot{P} = kP$ model of Example 2.1.

3. Linear Equations

Definition 3.1. A first-order equation is **linear** if it can be written in the *standard form*

$$\frac{dy}{dx} + P(x)\,y = Q(x),$$

i.e. y and y' appear only to the first power and are not multiplied together.

Why the method works. We look for an **integrating factor** $\mu(x)$ that turns the left-hand side into a single exact derivative. Multiplying through by μ,

$$\mu y' + \mu P y = \mu Q.$$

We want the left side to equal $(\mu y)' = \mu y' + \mu' y$. Matching the y-terms forces $\mu' = \mu P$, a separable equation for μ whose solution is

$$\boxed{\mu(x) = e^{\int P(x)\,dx}}$$

With this μ, the equation collapses to $(\mu y)' = \mu Q$, which integrates directly.

Recipe

(1) Put the equation in standard form and read off $P(x)$. (2) Form $\mu = e^{\int P\,dx}$. (3) The equation becomes $(\mu y)' = \mu Q$. (4) Integrate: $y = \dfrac{1}{\mu}\left(\displaystyle\int \mu Q\,dx + C\right)$.

Example 3.1. Solve $\dfrac{dy}{dx} + 2y = e^{-x}$. Here $P = 2$, so $\mu = e^{2x}$. Then

$$(e^{2x}y)' = e^{2x}e^{-x} = e^{x} \Rightarrow e^{2x}y = e^{x} + C \Rightarrow y = e^{-x} + Ce^{-2x}.$$

The term e^{-x} is the forced response; Ce^{-2x} is the decaying transient.

Where this shows up

- **Electrical circuits:** an RL circuit obeys $L\,\dot{I} + RI = E(t)$; an RC circuit obeys $R\dot{q} + q/C = E(t)$. Both are linear — the integrating factor gives charge and current for any driving voltage.
- **Mixing / brine tanks:** salt concentration in a tank with inflow and outflow, $\dot{S} + \frac{r}{V}S = r\,c_{\text{in}}$.
- **Single-compartment pharmacokinetics:** drug amount under infusion and elimination, $\dot{A} = R_{\text{in}} - kA$ — dosing schedules are designed from this.
- **Falling body with linear drag:** $\dot{v} = g - \frac{c}{m}v$ gives the approach to terminal velocity.

4. Exact Equations

Definition 4.1. Written in differential form $M(x,y)\,dx + N(x,y)\,dy = 0$, the equation is **exact** if there is a function $F(x,y)$ with $F_x = M$ and $F_y = N$. Its solutions are the level curves $F(x,y) = C$.

Theorem 4.1 (Test for exactness)**.** *On a simply connected region, $M\,dx + N\,dy = 0$ is exact if and only if*

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Why the method works. If such an F exists, the equation says $dF = F_x\,dx + F_y\,dy = 0$, so F is constant along solutions. Equality of mixed partials, $F_{xy} = F_{yx}$, forces the test $M_y = N_x$; conversely that test guarantees F can be reconstructed by integrating M in x and fixing the leftover y-terms via N.

Recipe

(1) Check $M_y = N_x$. (2) Set $F = \int M\,dx + g(y)$ (integrate M holding y fixed). (3) Differentiate in y and match to N to find $g'(y)$; integrate for $g(y)$. (4) Solution: $F(x,y) = C$.

Example 4.1. Solve $(2xy + 3)\,dx + (x^2 - 1)\,dy = 0$. Test: $M_y = 2x = N_x$ — exact. Integrate M in x: $F = x^2y + 3x + g(y)$. Then $F_y = x^2 + g'(y)$ must equal $N = x^2 - 1$, so $g'(y) = -1$, $g(y) = -y$. Hence

$$x^2y + 3x - y = C.$$

4.1 Special integrating factors

When $M_y \neq N_x$, the equation is not exact — but sometimes a factor μ makes it so. Two standard cases:

$$\frac{M_y - N_x}{N} = \text{function of } x \text{ only} \;\Rightarrow\; \mu(x) = e^{\int \frac{M_y - N_x}{N}\,dx},$$
$$\frac{N_x - M_y}{M} = \text{function of } y \text{ only} \;\Rightarrow\; \mu(y) = e^{\int \frac{N_x - M_y}{M}\,dy}.$$

Example 4.2. $(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0$ has $M_y = 3x + 2y$, $N_x = 2x + y$, not exact. But $\dfrac{M_y - N_x}{N} = \dfrac{x + y}{x(x + y)} = \dfrac{1}{x}$, so $\mu = e^{\int dx/x} = x$. Multiplying gives $(3x^2y + xy^2)dx + (x^3 + x^2y)dy = 0$, now exact, with solution $x^3y + \frac{1}{2}x^2y^2 = C$.

Where this shows up

- **Thermodynamics:** state functions (internal energy, entropy) have *exact* differentials; heat and work do not. Exactness is precisely what distinguishes a state function from a path-dependent quantity.
- **Conservative force fields:** a force is conservative exactly when its work form is exact, and then F is the potential energy.
- **Orthogonal trajectories & potential theory:** level curves $F = C$ describe equipotentials in electrostatics and steady heat flow.

5. Homogeneous Equations

Definition 5.1. $\dfrac{dy}{dx} = f(x,y)$ is **homogeneous** (of degree zero) if the right-hand side depends only on the ratio y/x:

$$\frac{dy}{dx} = F\Big(\frac{y}{x}\Big).$$

Equivalently, $M\,dx + N\,dy = 0$ with M, N homogeneous of the same degree.

Why the method works. The substitution $v = \dfrac{y}{x}$, i.e. $y = vx$, turns a ratio-only right side into something separable. Differentiating, $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$, so the equation becomes

$$v + x\frac{dv}{dx} = F(v) \implies x\frac{dv}{dx} = F(v) - v,$$

which *is* separable in v and x. Solve, then substitute back $v = y/x$.

Recipe

(1) Confirm the right side depends only on y/x. (2) Substitute $y = vx$, $y' = v + xv'$. (3) Separate and integrate in v, x. (4) Replace v by y/x.

Example 5.1. Solve $\dfrac{dy}{dx} = \dfrac{y^2 + 2xy}{x^2} = \Big(\dfrac{y}{x}\Big)^2 + 2\dfrac{y}{x}$. With $y = vx$:

$$v + xv' = v^2 + 2v \;\Rightarrow\; xv' = v^2 + v = v(v+1) \;\Rightarrow\; \int \frac{dv}{v(v+1)} = \int \frac{dx}{x}.$$

Partial fractions $\left(\frac{1}{v} - \frac{1}{v+1}\right)$ give $\ln\left|\dfrac{v}{v+1}\right| = \ln|x| + c$, so $\dfrac{v}{v+1} = Ax$. Back-substituting $v = y/x$: $\dfrac{y}{x+y} = Ax$, i.e. $y = Ax(x+y)$.

Where this shows up

- **Scale-invariant geometry:** curves whose slope depends only on *direction* from the origin — the defining property of many spirals and self-similar shapes.
- **Orthogonal trajectories:** families of curves cutting a given family at right angles often reduce to a homogeneous equation.
- **Self-similar phenomena:** problems with no intrinsic length scale (certain fluid, diffusion and boundary-layer problems) reduce to homogeneous form.

5.1 Aside: equations reducible by a linear substitution

An equation $\dfrac{dy}{dx} = f(ax + by + c)$ becomes separable under $u = ax + by + c$. For instance $\dfrac{dy}{dx} = (x+y)^2$ with $u = x + y$ gives $u' = 1 + u^2$, so $\arctan u = x + C$ and $y = \tan(x + C) - x$.

6. Reducible Equations: Bernoulli and Riccati

Some nonlinear equations are not solvable as they stand but become one of the earlier types after a clever substitution. Two are standard enough to name.

6.1 Bernoulli equations

Definition 6.1. A **Bernoulli equation** has the form

$$\frac{dy}{dx} + P(x)\,y = Q(x)\,y^n, \qquad n \neq 0, 1.$$

Why the method works. Dividing by y^n and setting $v = y^{1-n}$ linearises it. Since $\dfrac{dv}{dx} = (1-n)y^{-n}\dfrac{dy}{dx}$, the substitution yields

$$\frac{dv}{dx} + (1-n)P(x)\,v = (1-n)Q(x),$$

a *linear* equation in v — solve it by the integrating factor of §3, then return to $y = v^{1/(1-n)}$.

Example 6.1 (The logistic equation — Bernoulli with $n = 2$)**.** The logistic model $\dfrac{dP}{dt} = kP\left(1 - \dfrac{P}{M}\right)$ rearranges to $\dfrac{dP}{dt} - kP = -\dfrac{k}{M}P^2$. With $v = P^{-1}$ this becomes the linear equation $\dfrac{dv}{dt} + kv = \dfrac{k}{M}$, whose integrating factor e^{kt} gives $v = \dfrac{1}{M} + Ce^{-kt}$. Inverting,

$$P(t) = \frac{M}{1 + Ae^{-kt}}, \qquad A = MC,$$

the S-shaped curve — the same formula students verify in the Lecture 1 warm-up, now *derived*.

Where Bernoulli shows up

- **Constrained population growth:** the logistic model above (fisheries, epidemics, tumour growth).
- **Fluid mechanics:** boundary-layer and thin-film equations of Bernoulli type.
- **Economic growth:** the Solow–Swan capital equation is Bernoulli and linearises the same way.

6.2 Riccati equations

Definition 6.2. A **Riccati equation** is the general quadratic-in-y first-order equation

$$\frac{dy}{dx} = P(x) + Q(x)\,y + R(x)\,y^2.$$

Why the method works. Unlike the others, a Riccati equation has *no* elementary solution in general. But if *one* particular solution $y_1(x)$ is known (often guessed), the substitution

$$y = y_1 + \frac{1}{v}$$

removes the quadratic term and leaves a *linear* equation in v. So one known solution unlocks the rest.

Recipe

(1) Find or guess a particular solution y_1. (2) Substitute $y = y_1 + 1/v$. (3) Solve the resulting linear equation for v. (4) Recover $y = y_1 + 1/v$.

Example 6.2. Solve $\dfrac{dy}{dx} + y^2 = \dfrac{2}{x^2}$. Try $y_1 = \dfrac{a}{x}$: then $-\dfrac{a}{x^2} + \dfrac{a^2}{x^2} = \dfrac{2}{x^2}$ needs $a^2 - a - 2 = 0$, so $a = 2$ and $y_1 = \dfrac{2}{x}$ works. Put $y = \dfrac{2}{x} + \dfrac{1}{v}$; the equation reduces to the linear

$$\frac{dv}{dx} - \frac{4}{x}\,v = 1, \qquad \mu = x^{-4}, \qquad x^{-4}v = -\frac{1}{3x^3} + C.$$

Thus $v = Cx^4 - \dfrac{x}{3}$ and $y = \dfrac{2}{x} + \dfrac{1}{Cx^4 - \frac{x}{3}}$.

Remark 6.1 (Bridge to Unit 2)**.** The further substitution $y = -\dfrac{u'}{R\,u}$ turns any Riccati equation into a *second-order linear* ODE for u. This is the doorway from Unit 1 into the linear theory of Unit 2 — and the reason Riccati equations sit at the heart of so much applied mathematics.

Where Riccati shows up

- **Optimal control:** the *matrix Riccati equation* determines the feedback gain in LQR control and the Kalman filter — the backbone of modern control, including model-predictive control.
- **Quantum mechanics & wave propagation:** the Riccati form of the Schrödinger and WKB equations.
- **Finance:** affine term-structure and stochastic-volatility models yield Riccati systems.

7. Recognition guide: "What kind of equation is this?"

Faced with a first-order equation, run down this list *in order* and stop at the first match — the earlier methods are simpler, so try them first.

1. **Separable?** Does $\dfrac{dy}{dx}$ factor as $g(x)\,h(y)$? $\to$ separate and integrate (§2).
2. **Linear?** Can you write $\dfrac{dy}{dx} + P(x)y = Q(x)$? $\to$ integrating factor $e^{\int P\,dx}$ (§3).
3. **Exact?** In form $M\,dx + N\,dy = 0$, is $M_y = N_x$? $\to$ reconstruct F, solution $F = C$ (§4). If not, try a special integrating factor.
4. **Homogeneous?** Does the right side depend only on y/x? $\to$ substitute $y = vx$ (§5).
5. **Bernoulli?** Form $y' + Py = Qy^n$? $\to$ substitute $v = y^{1-n}$ (§6.1).
6. **Riccati?** Form $y' = P + Qy + Ry^2$? $\to$ find one y_1, substitute $y = y_1 + 1/v$ (§6.2).

Note that a single equation may pass several tests — e.g. $\dfrac{dy}{dx} + \dfrac{1}{x}y = 1$ is both linear and, after rearrangement, separable. Choosing the easiest applicable method is itself a skill this course develops (outcome H2).

Summary of standard forms

Type	Standard form	Key step
Separable	$y' = g(x)h(y)$	$\int dy/h = \int g\,dx$
Linear	$y' + P(x)y = Q(x)$	$\mu = e^{\int P\,dx}$, then $(\mu y)' = \mu Q$
Exact	$M\,dx + N\,dy = 0,\ M_y = N_x$	find F with $F_x = M$, $F_y = N$
Homogeneous	$y' = F(y/x)$	put $y = vx$
Bernoulli	$y' + Py = Qy^n$	put $v = y^{1-n}$
Riccati	$y' = P + Qy + Ry^2$	known y_1, put $y = y_1 + 1/v$

8. Self-check problem set

Not graded. Work these before the first tutorial; solutions to selected items follow. Identify the type first, then solve.

A. Classify and solve.

1. $\dfrac{dy}{dx} = \dfrac{x^2}{y}$
2. $\dfrac{dy}{dx} + 3y = 6$
3. $(2x + y)\,dx + (x - 2y)\,dy = 0$
4. $\dfrac{dy}{dx} = \dfrac{x + y}{x}$
5. $\dfrac{dy}{dx} + \dfrac{1}{x}\,y = x\,y^2$
6. $x\,\dfrac{dy}{dx} + y = y^2 \ln x$

B. Applied.

7. A tank holds 100 L of pure water. Brine with 2 g/L enters at 5 L/min while the well-mixed solution leaves at 5 L/min. Set up and solve for the salt content $S(t)$. *(Linear.)*
8. A cup of coffee at 85°C sits in a 22°C room and cools to 60°C in 10 min. Using Newton's law of cooling, find its temperature after 20 min. *(Separable / linear.)*
9. A population follows $\dfrac{dP}{dt} = 0.4\,P\left(1 - \dfrac{P}{5000}\right)$ with $P(0) = 500$. Find $P(t)$ and the time to reach 2500. *(Bernoulli / logistic.)*

Selected answers. **1.** $y^2 = \frac{2}{3}x^3 + C$ (separable). **2.** $y = 2 + Ce^{-3x}$ (linear). **3.** $x^2 + xy - y^2 = C$ (exact: $M_y = 1 = N_x$). **4.** $y = x(\ln|x| + C)$ (homogeneous *and* linear). **5.** Bernoulli ($n = 2$), $v = y^{-1}$ gives $v' - \frac{1}{x}v = -x$; hence $\dfrac{1}{y} = Cx - x^2$. **9.** $P(t) = \dfrac{5000}{1 + 9e^{-0.4t}}$; reaches 2500 when $t = \dfrac{\ln 9}{0.4} \approx 5.49$.